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TOPICAL REVIEW

A Systematic Review of Pretrained Models in Automated Essay Scoring

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ABSTRACT Automated essay scoring (AES) uses artificial intelligence and machine learning methods to grade student essays and produce human-like scores. AES research has witnessed significant advancements over time by adopting diverse machine learning models. This evolution started with traditional techniques like regression and classification, and later advanced to deep learning models that leverage neural networks for enhancing scoring performance. This review focuses on the utilization of Transformer architectures, a sophisticated form of neural networks employing attention mechanisms, with an emphasis on pretrained models like BERT in AES research. The aim is to enhance the understanding of their applicability in advancing the AES research landscape. Additionally, this study selected and analyzed relevant papers from Scopus and Web of Science databases in the past five years, by adhering to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. An intensive screening process was followed to shortlist studies within the defined scope, focusing on AES applications that employed pretrained models. From the comprehensive analysis of 354 studies, this review shortlisted 22 key papers and identified five distinct focus areas within the current AES research landscape: holistic scoring, trait scoring, cross-domain and generalization, model fairness, and robustness. The results highlight the potential of transformer-based pretrained models in improving AES systems' accuracy. However, pretrained models face several limitations, such as high computational costs, long input text length, and explainability. Tackling these challenges and integrating emerging technologies, such as large language models (GPT4), is expected to foster the development of accurate, robust, and transparent AES systems.

INDEX TERMS Automated essay scoring, PRISMA, trait, transformers, pretrained, BERT.

I. INTRODUCTION

A. AUTOMATED ESSAY SCORING OVERVIEW

Automated essay scoring (AES) is the process of using machines to grade students' essays automatically using machine learning (ML) and deep learning (DL) methods [1], [2]. AES is also called automated essay grading, automatic writing grading, and automatic essays. In 2003, [3] defined AES as the process of evaluating and assessing written work using computer programs. In addition, [4] stated that AES implies the grading of written text emphasized by prompt-based scoring against human benchmarks, while [5] viewed AES as a quality-driven regression task. Furthermore, [6] considered AES a specialized online technique to evaluate and score written essays by mimicking human intelligence. [7] believed that the effectiveness of an AES system relies on its ability to assign scores that closely resemble those provided by human graders. Achieving alignment between machine and human scoring is an essential benchmark for evaluating the effectiveness of AES systems. Therefore, AES can be summarized as the application of ML/DL models to analyze and grade student essays electronically. These models aim to provide consistent and objective scoring that could mimic human expertise.

Current automated assessment systems excel at grading objective questions, such as multiple-choice and true/false

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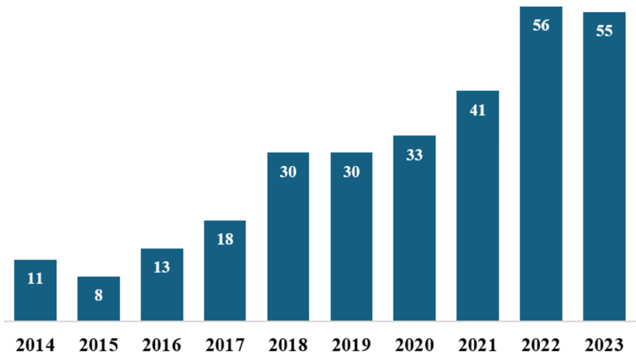

FIGURE 1. **Number of AES articles published between 2014–2023 (Scopus). The figure highlights the growing research interest in AES over the last decade.**

questions; however, they face challenges when grading short answers and essay questions. This challenge arises because answering open-ended questions requires processing complex natural language with a deep understanding of language structure and external domain knowledge, as [8] stated. The authors also highlighted that the answers to such questions might range from brief phrases to multi-paragraph essays, which poses a challenge for scoring systems because of the absence of a single or definitive answer.

Educators often rely on written essays to evaluate learner understanding and comprehension. Still, manual grading is labor-intensive, inconsistent, and time-consuming, especially with the rising student-to-teacher ratio, as noted by [9]. Scholars in [3] have proposed AES systems as a solution to these challenges. AES systems offer speed and consistency, addressing issues such as workload constraints that hinder educators from providing personalized feedback to students [10]. However, the authors in [11] highlighted AES limitations, including the lack of emotional understanding, less objectivity, algorithmic biases, and difficulty recognizing creativity, originality, and diverse writing styles. Despite these concerns, AES can alleviate some of the burdens on educators, enabling them to focus more on personalized guidance.

Interest in AES research gained a lot of attention over the last decade, as evidenced by the sharp increase in the number of published studies in Scopus (Figure 1). These figures were gathered by using the keyword "Automated Essay Scoring" in the Scopus search, with results grouped by year. This surge in AES studies reflects its potential to address key challenges educators face, such as producing accurate scores.

B. TRANSFORMER ARCHITECTURE AND PRETRAINED MODELS

The field of NLP witnessed a huge transformation with the advent of DL models. A key development within DL has been the introduction of the Transformers architecture, which utilizes attention mechanisms [12]. Since then, it has become a fundamental model in NLP tasks such as text

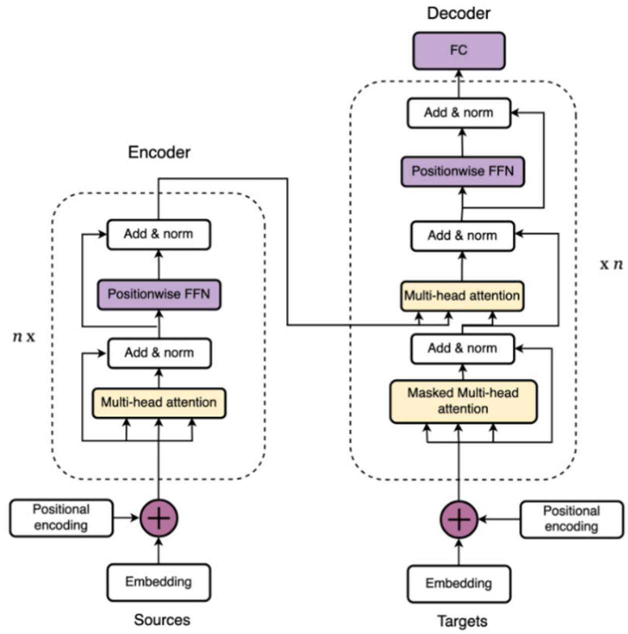

FIGURE 2. **Transformer encoder-decoder architecture [12].**

classification, summarization, translation, and text generation. References [12], [13], and [14] have listed the key features of the Transformers architecture as follows:

- The original Transformers architecture consists of an encoder and decoder structure. The encoder processes the input data and generates a set of representations, while the decoder uses these representations to generate the output sequences. Both the encoder and the decoder consist of multiple layers that include attention heads and feed-forward neural networks. The transformer encoder-decoder architecture is shown in Figure 2.
- It utilizes self-attention mechanisms to allow the model to weigh the important words in a sentence and capture the long-range dependencies in the data more effectively than CNNs and RNNs.
- Transformers can process the data in parallel, unlike RNNs, which process data sequentially.
- Since Transformers don't process data sequentially, they don't inherently store the order of words. Hence, they employ positional encoding to inject information about the position of the words in the sequence.

The innovative Transformer architecture paved the way for the development of pretrained models. These powerful models have been trained on massive datasets and leverage transfer learning mechanisms, where the knowledge gained while solving one problem is applied to similar problems in the future. In the context of NLP, pretrained models have significantly advanced the field with the ability to understand contextual information and excel in various tasks, such as text classification, summarization, translation, and text generation. Some famous examples of Transformer-based pretrained models include Bidirectional Encoder Representations from Transformers (BERT), Generative Pretrained Transformer (GPT), and eXtreme Language Model Pretraining Network (XLNET). These advanced models can also be fine-tuned for specific tasks with relatively smaller datasets to improve the model's accuracy [15], [16], [17].

Transformer-based models have become widely used across research disciplines due to their ability to model sequential and high-dimensional data. Their applications have been expanded to many areas other than NLP. For example, [18] employed a Transformer-based architecture that utilized GPT-2 for time-series analysis to predict particle trajectories. The model was integrated with a 3D CNN to enhance surface erosion predictions in dynamic systems. In the field of quantum computing, [19] introduced a variational quantum circuit architecture (SASQuaTCh) that mimics the self-attention mechanism of classical transformers, enabling efficient processing of spatiotemporal data. In healthcare, [20] applied the Vision Transformer (ViT) to distinguish COVID-19 infections by analysing chest X-rays with high accuracy. Additionally, in the financial sector, [21] utilized transformer-based models to predict loan defaults, leveraging self-attention to capture complex dependencies among financial indicators. In the education field and AES in particular, various Transformer-based pretrained models have been employed, such as BERT and its variants (DistilBERT, RoBERTa, and Sentence BERT). Additionally, Longformers, a specialized Transformer architecture designed for processing long text sequences, has been implemented to overcome the limitations associated with handling lengthy essays. As AES is the primary focus of this study, relevant Transformer-based AES research will be reviewed in detail in the upcoming sections.

C. RESEARCH OBJECTIVES

This study conducted a systematic analysis of AES research between 2019 and 2023, with a focus on the utilization of Transformer-based pretrained models across different AES tasks. The selected time frame is significant because the Transformers architecture was introduced in 2017 [12], making it challenging to find relevant research papers before 2019. After 2023, the focus of NLP research shifted toward the Large Language Models (LLMs), which makes later studies less representative of Transformers-focused AES work. Earlier studies on AES systems based on traditional ML and NN models are briefly reviewed but excluded from the shortlisted papers. In addition, this study attempts to provide a comprehensive overview of the AES research landscape, including its development history, available datasets, and identification of current limitations.

The research questions addressed in this study are as follows:

- **RQ1:** How has AES research evolved over time, and what publicly available datasets are most commonly used for AES tasks?
- **RQ2:** How have advanced deep learning models, particularly transformer-based pretrained models, been utilized to address various AES tasks in research between 2019 and 2023?
- **RQ3:** What are the key challenges and limitations in AES research, with a particular focus on transformer-based pretrained models?
- **RQ4:** What future directions and innovative approaches hold potential for improving the performance and generalization of AES systems?

The remainder of this paper is organized to provide a clear overview of the selection and review process followed to obtain the results. Section II covers the previous literature review studies done within the same research context, highlighting existing knowledge and identifying potential research gaps. Section III introduces PRISMA and its guidelines. Section IV details the selection process, outlining the shortlisted papers and the key results obtained from the systematic review. Section V presents a thorough discussion of the findings, discussing their significance and limitations. Lastly, section VI concludes the study with recommendations and future research directions.

II. LITERATURE REVIEW

A. RELATED WORK

A search on Scopus[1] and Web of Science (WoS)[2] databases reveals recent literature review studies on AES and related topics, published over the last five years. These reviews vary in depth, with some of them offering comprehensive insights into AES development, methodology, and contributions. This section discusses the key studies that have significantly contributed to the understanding and advancement of AES research.

The authors in [22] studied and classified AES systems based on feature extraction into handcrafted and automated systems, primarily focusing on AES systems for English essays. In addition, they identified the limitations of AES systems compared to human raters, such as AES systems can be deceived easily, and the lack of sensing creativity. However, the authors overlooked newer techniques, such as Transformers, in their review. Similarly, [23] categorized AES systems into three frameworks: content similarity, ML, and hybrid approaches. However, they focused on traditional regression and classification techniques and did not address newer advancements such as DL or Transformers. They also highlighted several AES limitations, such as no one-size-fits-all solution, prompt dependency, the nonlinear relationship between essay features and grades, and the need for effective feedback mechanisms.

The authors in [24] classified AES systems into handcrafted features, DL, and Transformer-based systems. However, they did not provide a technical overview of AES systems and only covered four Transformer-based studies.

[1] https://www.scopus.com
[2] https://mjl.clarivate.com/search-results

They advocated for the continued use of handcrafted feature approaches along with fine-tuning the new techniques. They stressed the importance of studies focusing on multi-trait and cross-prompt scoring. Whereas [25] concentrated on deep neural network (DNN) based AES systems with limited coverage of traditional approaches and no comparative performance analysis. They also emphasized the need for more efficient models that cover trait scoring and cross-prompt approaches.

The authors in [9] conducted a comprehensive review of AES studies conducted between 2010 and 2020. They identified various challenges, including the need for domain-specific datasets, limitations in assessing essay completeness, challenges in providing feedback, and concerns regarding model robustness against adversarial attacks and cheaters. Additionally, [26] observed fluctuations in AES research over the past two decades. While it rose between 2009 and 2013, it dropped in 2014 and 2015 because it was difficult to address research gaps with available traditional ML techniques, such as latent semantic analysis (LSA). However, a resurgence began in 2016, likely due to the adoption of new ML techniques, such as DL models. However, this review did not compare the performances of different models. Another thorough study by [27] covered AES studies between 2019 and 2021, highlighting the limitations of AES systems, such as grading math and equation questions, in addition to extended essays.

Previous AES review studies have either focused on traditional ML models that employed handcrafted features [22], [23], or the models that leveraged DNN techniques [25]. However, to the best of our knowledge, a few reviews covered the Transformers-based pretrained models. Examples that have touched on this area include the review studies [24], [27]. The current study attempts to fill this gap by providing a comprehensive systematic review of studies that used the Transformers architecture, specifically pretrained models, in various AES tasks. Thus, this study aims to enhance understanding of the effectiveness and applicability of Transformer-based pretrained models in advancing the AES research landscape. In addition, none of the aforementioned reviews applied the PRISMA methodology for conducting systematic literature reviews. The current study adapts the PRISMA guidelines to ensure transparency, reduce bias, and foster a more comprehensive understanding of the AES research landscape.

B. OVERVIEW OF AES RESEARCH

Before delving into the research methodology and presenting the findings of this systematic review, it is essential to discuss some key topics in AES research, including its history, development, classification, and available datasets. This discussion provides the necessary context to better understand the results obtained from the shortlisted papers and contributes to addressing Research Question 1 (RQ1).

TABLE 1. Summary of AES studies utilized statistical models.

Ref.	ML Model	Dataset	Metric	Application
[6]	Classification / Random Forest	Custom	Acc: 0.97	Evaluated the performance of different classification approaches. Random Forest outperformed other models.
[30]	Regression / MLR	Custom	Acc: 0.77	Focused on two types of features: syntax and semantics. Utilized MLR model with the fusion of fuzzy ontology and LSA.
[31]	Classification / Random Forest	ASAP	QWK: 0.74	Used Random Forest classifier to score each essay attribute individually.

1) AES HISTORY AND DEVELOPMENT

AES has a well-established research history that spans several decades, marked by technological advancements. The journey began with the Project Essay Grader (PEG), introduced in 1960, as the first AES system that predicted the basic text quality using correlation coefficients. Over the years, AES research continued to evolve as well and various commercial AES products have emerged. Notable among these commercial products is the Intelligent Essay Assessor (IEA),[3] introduced in 1998, which used Latent Semantic Analysis (LSA), and was later acquired by Pearson in 2004. The e-rater[4] and Criterion[5] were developed by Educational Testing Services, employed statistical models and NLP to analyze linguistic features, while IntelliMetric,[6] introduced by Vantage Labs, was the first AI-powered AES system combining AI, NLP, and statistical models. Furthermore, another key development is the Enhanced AI Scoring Engine (EASE),[7] an open-source scoring engine developed by ASAP competition winners, which is frequently used as a benchmark in AES research [9].

Initially, AES systems relied heavily on statistical models that required manual feature engineering, which is time-consuming and limited by the quality of the extracted features [9], [22]. Despite their early success, these models struggled with complex aspects like cohesion and coherence, as well as semantic relationships, because they primarily focused on shallow linguistic features. However, these traditional approaches have the advantage of strong explanatory power, making it easier to understand how included features contributed to the final score [25], [28], [29]. Table 1 lists some examples of studies that utilized statistical models in AES tasks.

The advent of DL marked a transformative shift in AES research as it eliminated the need for manual feature engineering. NN-based models such as CNNs and RNNs models, particularly LSTM, outperformed traditional

[3]https://www.pearsonassessments.com/large-scale-assessments.html
[4]https://www.ets.org/erater.html
[5]https://www.ets.org/criterion/about.html
[6]https://www.intellimetric.com/direc
[7]https://github.com/openedx-unsupported/ease

TABLE 2. Summary of AES studies utilized neural network models.

Ref.	NN Model	Dataset	QWK	Application
[1]	CNN + RNN / LSTM	ASAP	0.708	The first NN-based model that utilizes both CNN and RNN. The model uses a lookup table with the one-hot representation of the word vector of an essay to generate the final score.
[32]	LSTM-CNN-Atten	ASAP	0.764	A model that uses attention pooling with CNN used for character and word embedding, then uses LSTM to generate the final score.
[33]	SkipFlow (BiLSTM)	ASAP	0.764	NN approaches that incorporate logical and semantic flow into the LSTM model.
[34]	LSTM + atten	ASAP / ASAP++	–	Used NN model for trait scoring of cross-prompt essays.
[35]	LSTM	ASAP / ASAP++	0.592 / 0.586	Used LSTM model (ProTACT) for cross-prompt essay trait scoring.

methods and improved scoring capabilities with their ability to capture textual features, understand context, and maintain semantic sequences [9], [25]. Furthermore, the development of advanced NN architectures, such as Transformers, opened new horizons in AES by further enhancing scoring capabilities. However, NN-based models also introduced new challenges, such as the need for large datasets and significant computational resources. Table 2 lists some key studies that utilized NNs in AES tasks.

Based on the above, AES can be classified into different eras that reflect the technological advancements of that time, such as concept mapping (1966), information extraction (2000), and corpus-based methods (2006), eventually leading to the ML era (2009), where AI-powered algorithms became prominent. Furthermore, online competitions, such as the ASAP hosted on Kaggle in 2012, further accelerated AES research progress by providing a platform for innovation [8]. In Addition, researchers have explored different scoring methods, starting with holistic scoring in which a single score is generated for the whole essay, and trait scoring in which different aspects are evaluated and then the final score is calculated based on these aspects. Recent trends have seen a growing interest in trait scoring, which evaluates different aspects of an essay to enhance model explainability, interoperability, and generalization. This evolution in scoring methods signifies a shift towards more nuanced and comprehensive approaches in AES research [25].

Although the need for manual feature engineering was reduced with the use of DL models, it remains important in specific cases, such as trait scoring. Combining handcrafted features with DL approaches could improve scoring accuracy. This evolution shows the ongoing development in AES research, aiming to address various aspects of essay scoring comprehensively [28], [36], [37].

2) DATASETS

Compiling a dataset for AES experimentation is a labor-intensive process that poses a major challenge. It involves various critical tasks, including prompt generation (questions), securing learner responses, engaging at least two raters for grading and feedback, and subsequent data cleaning and preprocessing. Due to this complexity, many researchers use publicly available datasets from competitions or other studies.

Among the available datasets, the ASAP dataset is the most widely used benchmark for comparing results across AES studies. It gained popularity after being featured in a Kaggle competition funded by the Hewlett Foundation in 2012. The dataset has about 13,000 essays (with 100-600 words each), scored by two expert graders. It includes eight tasks, with responses from students in grades 7, 8, and 10. The tasks are divided into persuasive, source-based, and opinion-based prompts, with five scored holistically and three using trait scoring [7], [38], [39]. Table 3 provides detailed information about the ASAP dataset.

In the ASAP dataset, most tasks were scored holistically. To address this limitation, the authors in [31] extended the dataset by introducing trait scores for all eight prompts, with the support of expert human annotators. They named the enhanced version ASAP++ and included traits such as content, organization, word choice, sentence fluency, and conventions. The authors proposed ASAP++ as a potential gold standard for future AES trait-scoring tasks.

Other datasets used in AES tasks include TOFEL11 [40], the Cambridge English assessment dataset, and the domain-specific Operating System dataset by [29]. Furthermore, many scholars such as [6] and [29] emphasized the need for a wider range of AES datasets, including domain-specific ones. This could potentially foster AES research and improve model generalization, especially scoring across unseen prompts. Table 4 lists the commonly used datasets in AES research, including their characteristics and access links.

3) EVALUATION METRICS

The performance of ML/DL models is evaluated using metrics depending on the task. Accuracy, precision, and recall are used for classification, while absolute error is used for regression. However, most AES studies prefer Cohen's Kappa with quadratic weights (QWK), because it accounts for the random agreement between the actual and predicted scores and minimizes bias from chance. In addition, QWK was the official evaluation metric in the ASAP competition [41].

Generally, the efficiency of an AES system is measured by the agreement between the human-assigned scores (gold standard) and machine-generated scores [7], [9], [38]. Typically, this agreement is evaluated using three key metrics:

- **Quadratic Weighted Kappa (QWK)**: Measures the agreement between human scores and machine-generated scores. It ranges between 0 and 1; a value of 1 implies perfect agreement, a value of 0 suggests

TABLE 3. Summary of ASAP essay tasks.

Task	Task Description	Records	Score Range	Avg Word	Grade	Task Type	Rubric
1	Convince a local newspaper reader on how people are affected by computers	1783	2–12	371	8	Persuasive	Holistic
2	Express opinions on whether certain media should be banned from libraries based on personal experience	1800	1–6	386	10	Persuasive	2 traits
3	Describe how the environment influences a narrator traveling by bicycle, given the provided text	1726	0–3	110	10	Source-based	Holistic
4	Explain why the text "Winter Hibiscus" ends in a particular way	1772	0–3	95	10	Source-based	Holistic
5	Describe the mood of a writer using a given text as a memoir	1805	0–4	123	8	Source-based	Holistic
6	Describe the obstacles faced in the construction of the Empire State Building based on a given text	1800	0–4	155	10	Source-based	Holistic
7	Write an article about patience	1596	0–30	170	7	Opinion - Expository	4 traits
8	Write a true story involving laughter	723	0–30	615	10	Opinion - Narrative	6 traits

TABLE 4. Summary of datasets used in AES research.

Dataset	Description	Records	URL
ASAP	Automated Student Assessment Prize dataset used for essay scoring. It contains 8 different essay prompts, each with multiple responses.	12,976	ASAP
ASAP++	An extended version of ASAP with further annotations, and trait scores for the 8 prompts.	12,976	ASAP++
TOEFL11	The TOEFL11 dataset is a large-scale corpus of English essays written by non-native speakers from 11 different native language backgrounds. Essays are labeled with proficiency levels and various linguistic features.	12,100	TOEFL11
Cambridge English Assessment	Dataset for evaluating English language proficiency. It contains records of exam scripts collected between 2016 and 2020. It has four traits, each rated out of 5, leading to an overall score of 20.	50,000	CLC FCE
CELA	Chinese English Foreign Language learners' argumentation dataset for assessing argumentation in essays written by Chinese learners. It has 6 traits, and the final score is the average of the two raters' scores.	300	CELA
CRP	The CommonLit Readability Prize dataset consists of excerpts from various periods, with reading ease scores ranging from -3.68 to 1.72. The average excerpt length is 175 words.	2,834	CRP
NAEP	The NAEP (National Assessment of Educational Progress) Challenge dataset contains student responses that range from one sentence to five sentences, leading to different scoring ranges.	18,000	NAEP
OS	The Operating System dataset focuses on essays related to operating systems. The responses were collected from college students based on five OS-related questions and evaluated by two experts.	2,390	OS

agreement by chance, and negative values indicate worse-than-chance agreement. It is an extension of Cohen's Kappa and is calculated using the formula:

$$W_{(i,j)} = \frac{(i-j)^2}{(R-1)^2}$$

where i is the human rating, j is the AES rating, and R is the total number of possible ratings [32].

- **Mean Squared Error (MSE)**: Measures the squared difference between human's scores and system-generated scores. It is typically used in regression tasks [34].
- **Distance-Based Measures**: Methods like cosine similarity and the Pearson correlation coefficient (PCC) are used to determine the distance between a student's response and a reference answer and assign a score accordingly. The PCC ranges from -1 to 1, where 0 indicates no relation, 1 signifies a perfect positive correlation, and -1 represents a perfect negative correlation between the two variables.

III. METHODOLOGY

A. THE FOUR STEPS OF AES DEVELOPMENT

Generally, the development of ML models follows a well-established workflow that comprises several stages. Interestingly, the development process of AES showed a notable alignment with this established paradigm. Reference [42] outlined the AES development process and divided it into four key steps, which encompass:

- **Preprocessing**: This initial step ensures that student responses are presented in a standardized electronic format. It typically involves cleaning and formatting the text, such as removing special characters and inconsistencies, and preparing the text for the subsequent steps.
- **Feature Extraction**: In this step, the student responses are transformed into a set of features that ML models can understand and utilize. With traditional ML models, this step is done manually by human experts. However, deep learning (DL) models have evolved as a powerful alternative, enabling automatic feature extraction from

the data. DL approaches offer greater efficiency, as they can capture complex patterns that might be missed by manual methods.
- **Model Building**: Here, the ML model is created by training it on student responses paired with the corresponding human rater scores. AES research has witnessed significant advancement over time with the adoption and refinement of various ML models.
- **Essay Scoring**: In this step, the trained model is deployed and put into action to score unseen essays that were not encountered during the training step. The output scores can take different forms based on the utilized model. Regression models are used to predict a single discrete value that might range between 0 and 1, or 0 and 100. Alternatively, classification models assign category labels, such as letter grades (A–F). After testing the model, its performance is evaluated by comparing the generated scores with human raters' scores to assess the model's accuracy.

B. OVERVIEW OF PRISMA

To ensure transparency and minimize bias, this systematic review adheres to PRISMA 2020[8] guidelines [43]. The PRISMA statement was originally published by [44] and was designed for healthcare systematic reviews. However, it can be applied to other disciplines, e.g., computer science studies. This comprehensive framework provides guidelines encompassing all stages of the research process, including crafting a clear and informative title and abstract, outlining detailed methods, presenting results, and drawing well-supported conclusions while avoiding potential bias. Adhering to the PRISMA guidelines enhances the clarity, transparency, and trustworthiness of the findings.

C. SEARCH SOURCES AND TERMS

A thorough search was conducted across two reputable academic databases, Scopus and WoS, to identify relevant research articles. These databases were selected for their extensive coverage of peer-reviewed articles to ensure the quality and credibility of the retrieved information. The search yielded 354 results, of which 177 were obtained from Scopus and 177 from WoS.

To ensure that the search results directly address the research scope and capture the most recent advancements in the field, the search terms were refined using keywords like "Automated Essay Scoring", "Automated Essay Grading," "BERT," "Transformers," and "pretrained models." In both Scopus and WoS databases. In addition, the results were limited to English-language journals published within the last five years and categorized under the "computer science" subject area. This focus on recent publications streamlines the review process and ensures that our analysis incorporates the latest developments in ML and DL models employed for AES tasks.

[8] http://www.prisma-statement.org

As mentioned earlier, this study focuses on the last five years, specifically the utilization of Transformer-based pretrained models in AES tasks. Since models like BERT, which was built on the Transformers architecture, emerged in 2018, earlier studies fall outside the scope and are considered less relevant.

D. INCLUSION AND EXCLUSION CRITERIA

This study adopts strict inclusion and exclusion criteria to ensure a comprehensive, high-quality review. Only research papers published between 2019 and 2023 in reputable, peer-reviewed journals are considered. The focus is on studies employing recent advancements in DL models in the AES context. Conversely, nonoriginal sources such as reviews, surveys, book chapters, presentations, editorials, commentaries, and non-English-language corpora studies are excluded. This focus on recent, high-quality, and relevant research guarantees a thorough analysis of current advancements in utilizing DL models for AES. The inclusion criteria included:

- Studies that address the use of ML/DL techniques, especially those utilizing Transformers-based pretrained models.
- Studies published within the last five years.
- Studies proposing novel approaches, focusing on key AES tasks, or contributing new knowledge to AES development.

While the exclusion criteria included:

- Non-peer-reviewed articles, including book chapters, reviews, surveys, or presentations.
- Studies that utilize non-English datasets.
- Studies with a different scope but related context, such as those focused on improving writing quality or evaluating commercial AES products.

E. ARTICLE SELECTION AND SCREENING

The selection of relevant studies was conducted in three phases. First, target databases were searched to obtain an initial pool of articles. The results were exported in (.ris) format and imported into Rayyan,[9] an online systematic review screening tool. At this stage, duplicate studies present in both datasets were identified and reviewed. One copy of each study was retained while the duplicates were removed to ensure accuracy and avoid redundancy. Literature reviews and other non-peer-reviewed articles were excluded, including book chapters, surveys, presentations, and incomplete work. In addition, studies that utilize non-English corpora were excluded. The titles and abstracts of the remaining studies were screened to identify relevant studies. Each study was screened, and relevant labels were added accordingly. Finally, a full-text reading of the shortlisted articles was performed to ensure that only high-quality and relevant studies were included. Studies that failed to satisfy the established criteria were excluded.

[9] https://new.rayyan.ai

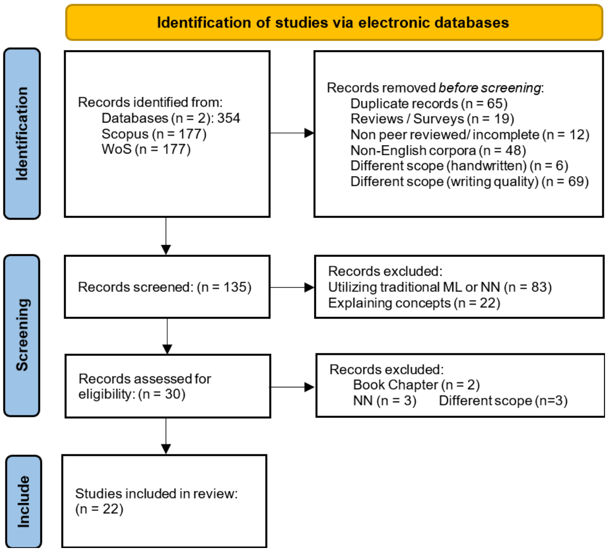

FIGURE 3. PRISMA flow diagram to identify relevant studies. The figure illustrates the inclusion and exclusion process, showing the number of studies at each screening stage.

Following the above selection criteria, the initial search of the Scopus and WoS databases yielded 354 citations, from which duplicates and non-relevant studies were excluded. The remaining articles were then screened by carefully reading the titles and abstracts of each article. This resulted in the exclusion of 324 studies. Eight additional citations were excluded after full-text reading and examination of the remaining 30 studies. Ultimately, 22 studies were considered eligible for inclusion in this systematic review. Figure 3 shows the PRISMA flow diagram outlining the process used to identify the relevant studies.

F. DATA SYNTHESIS

During the full-text reading, tags were assigned to categorize the shortlisted papers and to identify the relevant focus areas. After completing this task, the analysis revealed five key themes, which are AES holistic scoring, trait scoring, cross-prompt generalization, model robustness against adversarial inputs, and model fairness.

G. QUALITY ASSESSMENT

To guarantee the quality and relevance of the included studies, we employed a rigorous evaluation process. Each study was carefully assessed against several key criteria. A major focus was put on the research originality, which involves assessing the novelty of the research question, whether addressing under-explored areas, and the comprehensiveness of the employed methodology. Additionally, the clarity and details of the reported results, along with the contribution to the AES research landscape, are also assessed. Furthermore, the appropriateness of the chosen ML/DL models and the utilized datasets were examined. By thoroughly assessing these factors, we ensure the validity and overall soundness of the included studies in this review.

TABLE 5. Summary of included studies by year and focus area.

Year of Publication	# of Studies
2020	4
2021	4
2022	5
2023	9
Focus Areas	**# of Studies**
Holistic Grading	15
Trait Scoring	5
Robustness	4
Cross-prompt	6
Fairness	3

H. BIAS ASSESSMENT AND AVOIDANCE

To ensure objectivity and to avoid Bias, two independent reviewers carried out the screening process. Each study was assessed against the predefined inclusion and exclusion criteria. Any discrepancies or disagreements in selection were resolved through discussion and consensus-building. In cases where disagreements persisted, additional steps were taken, such as considering the seniority of the team members, leveraging domain expertise, and consulting external experts, if needed. This collaborative and informed decision-making process ultimately resulted in shortlisting the 22 papers that met the research objectives and were agreed upon by both reviewers.

IV. RESULTS

This section presents the key findings from the shortlisted studies in the systematic review. The subsections below highlight how Transformer-based pretrained models have been applied in AES research, which contributes to addressing Research Question 2 (RQ2).

A. CHARACTERISTICS OF INCLUDED STUDIES

The statistics of the included studies are presented in Table 5. As can be seen, the number of AES studies utilizing Transformers pretrained models is increasing. Additionally, the depth of these studies has become increasingly sophisticated. For example, in 2020, the authors primarily considered whether to fine-tune pretrained models. In 2023, the authors incorporated pretrained models into hierarchical architectures with multiple layers using various advanced ML models, e.g., Transformers, DNNs, and CNNs. The studies included in this systematic review and their corresponding applications are summarized in Table 6.

B. FINDINGS OF INCLUDED STUDIES

1) HOLISTIC SCORING

Holistic scoring is a type of evaluation that assesses the essay by providing a single overall score, reflecting the quality of the essay as a whole, without a breakdown of how that score was determined. In AES, holistic scoring involves training the ML model to generate an overall score by mimicking

TABLE 6. Summary of selected studies utilizing pretrained models for AES.

#	Ref	Model	Dataset	Evaluation Metrics	Application
1	[5]	BERT	ASAP	QWK	Fine-tuned R2BERT model using a multi-loss strategy that combined regression and ranking to enhance AES holistic scoring performance.
2	[16]	BERT	ASAP	QWK	Examined if fine-tuning pretrained model affects the scoring performance.
3	[17]	BERT	ASAP (P8), CRP	QWK	Improved BERT performance by employing multiple losses and transfer learning to enhance cross-prompt scoring.
4	[28]	SBERT	ASAP	QWK	Proposed a multi-scale hybrid model that utilized SBERT for sentence vectors and LSTM-MoT for document-scale features to improve essays holistic scoring.
5	[29]	SBERT + LSTM	ASAP, OS	QWK	Used a multi-task model that utilized SBERT for essay features extraction and LSTM for sentence processing to generate overall essay and trait scores.
6	[36]	BERT	ASAP	QWK	Suggested combining manual features with deep encoded features to improve AES holistic scoring performance.
7	[37]	BERT / Longformer	ASAP	QWK	Used BERT and Longformer models to improve AES scoring by evaluating the grammatical and stylistic traits to assess the overall essay quality.
8	[42]	BERT	ASAP	QWK	Fine-tuned BERT using three Active Learning strategies to reduce the number of manually scored essays required to build accurate AES systems.
9	[45]	GloVe / ELMo / BERT	ASAP	QWK	Compared various pretrained models' robustness against adversarial essays.
10	[46]	DistilBERT	Cambridge English dataset		proposed a multi-task learning (MTL) NN model built on top of DistilBERT for holistic scoring.
11	[47]	BERT	ASAP, CELA	QWK	A hierarchical model that utilized BERT with MTL for both holistic and trait scoring.
12	[48]	BiLSTM / RoBERTa	ASAP	QWK	Combined Bi-LSTM model over pretrained RoBERTa model to address the coherency issue. Used Bi-LSTM to address long essays and included the prompt information in the input text to improve holistic essay scoring.
13	[49]	CapsNet / BERT	ASAP	QWK	Used CapsNet with BERT contextualization model to detect adversarial essays and score them lower.
14	[50]	BOW / TFIDF / SBERT / LASER	ASAP (P7)	QWK	Compared different models performance on scoring fairness and their ability to detect paraphrased essays.
15	[51]	BERT	ASAP	QWK	Utilize BERT to reduce the number of trainable parameters using smaller datasets.
16	[52]	SBERT	ASAP	QWK	Compared fine-tuned SBERT vs. non-fine-tuned. PCA used with SBERT to identify scoring-relevant features.
17	[53]	In-context-BERT	NAEP	QWK	Employed a fine-tuned in-context BERT to score reading comprehension questions by generating a single holistic score while providing contextualized information on each item.
18	[54]	SBERT	ASAP	Accuracy/ Recall/ F1	Used SBERT to detect off-topic essays and score them lower using cosine similarity.
19	[55]	BERT variants	ASAP	Accuracy	Used Transfer Learning with multiple BERT variants to generalize to low-data topics. Also added essay subjects to the training dataset to capture essay-topic relationships. For data augmentation, they utilized BART to insert a summary of the topic every 10 lines, thereby maintaining topic focus, especially with long essays.
20	[56]	BERT + NN	ASAP, TOEFL11	QWK	Proposed PANN, prompt-aware neural network model designed to improve scoring by jointly learning prompt adherence and overall writing quality features. It leverages BERT in the EQ-net for capturing essay quality features and an interaction-based PA-net for modeling prompt-specific relevance.
21	[57]	Longformer		QWK	Used Longformers for long text handling; trial for model interoperability and explainability.
22	[58]	BERT	ASAP (P1,2,7)	QWK	Introduced Topic-aware BERT, a model that predicts holistic essay scores and provides meaningful feedback by identifying key topical sentences using self-attention analysis.

the holistic evaluation process employed by humans to grade written essays. This task is the most common in AES research due to its simplicity. To better understand this research area, Table 7 summarizes some shortlisted studies that utilized Transformers-based pretrained models to generate holistic essay scores.

The primary focus of the listed studies is to enhance the model performance and achieve the highest possible QWK. It was noticed that BERT emerged as the most frequently used model, with the ASAP dataset being the most common benchmark across these studies. Notably, [47] reported achieving the highest QWK of 0.83. It was also noticed that most studies didn't rely solely on pretrained models alone as they may fail to deliver satisfactory results. Instead, a combination of models and different learning techniques were employed to improve scoring accuracy. Additionally, long text handling posed a challenge, which [48] addressed by employing a BiLSTM model alongside the pretrained model.

Although the Longformer model offers another solution for managing long texts, none of the listed studies utilized this approach.

2) TRAIT SCORING

Rather than grading the essay holistically by producing a single overall score, trait scoring examines deeper scoring based on an essay's strengths and weaknesses. This technique attempts to evaluate the given essay based on multiple aspects that are frequently referred to as traits. These traits can represent various aspects of writing quality, such as adherence to prompts (the student's response is focused on the assigned topic), content and development, grammar and vocabulary, style, and organization.

Many studies have attempted to explore AES systems that support trait scoring because they offer several advantages. For example, trait scoring can help provide students with useful feedback [29]. In addition, the authors in [57] argued that trait scoring can enhance the overall transparency and explainability of the score generated by the model. Many scholars have employed Transformers-based pretrained models in exploring trait scoring in AES research. Reference [47] introduced a hierarchical model that incorporates Multi-Task Learning (MTL), a fine-tuned BERT, and attention mechanisms to effectively process long essays addressing the inherent 512-token limitation of BERT. The attention mechanism enabled the model to identify the key textual parts that influence scoring, and which in turn enhanced the overall scoring accuracy. When evaluated on the CELA dataset, the model achieved strong QWK across individual writing traits, including Grammar (0.87), Lexicon (0.80), Global Organization (0.79), Local Organization (0.72), and Supporting Ideas (0.74). In the same context, [29] utilized SBERT with LSTM to capture coherence, particularly traits like organization and word choice. The proposed model outperformed baselines with a QWK of 0.67 for both traits. Additionally, [57] employed Longformer to address the long essay challenge. The model was designed to assess various writing traits and demonstrated strong performance while achieving QWK scores of 0.78 for Main Idea, 0.84 for Language, and showing correlations of 0.44 with Word Count and 0.55 for Support. Table 8 presents examples from the shortlisted studies that focused on trait-level scoring.

However, challenges also exist. One challenge is related to model complexity when handling multiple traits that must be understood, evaluated, and integrated into the scoring process. Another challenge is the need for annotated datasets in which essays are graded holistically, along with various traits. Some available datasets used in trait scoring include ASAP tasks 2, 7, and 8, ASAP++, and TOFEL11 datasets.

3) GENERALIZATION/CROSS PROMPT

A key capability for any ML/DL model is the ability to generalize across unseen data from different domains and provide accurate results, which is commonly referred to as cross-domain generalization. Typically, this process can be enhanced using large and diverse datasets to allow the model to capture more generalizable patterns. Enhancing model generalization is a well-established topic in AES research, as shown by the studies detailed in Table 9. However, this area of research has a limited number of studies with limited progress. The following methods are commonly used to investigate generalization when utilizing the ASAP dataset.

- **Same Genre – Cross Prompt**
  This method evaluates the model's ability to generalize across similar but not identical tasks. The model is trained on one specific prompt and then tested on a different prompt within the same genre, such as using two different opinion prompts. Reference [16] argued that ASAP prompts 2–6 are more generalizable to the AES context as they are scored with 4–6 point scales. While each prompt has unique characteristics, such as differences in student age and question types, these differences should not influence the model's performance. A robust AES model should be able to adapt to these variations with no degradation in accuracy.
- **Cross Genre – Cross Prompt**
  This method evaluates the model's ability to generalize across different genres. The model is trained on a prompt from one genre and then tested on a prompt from a different genre. For example, the model could be trained using an opinion prompt and then tested on an argumentative prompt. This approach is particularly challenging because it requires the model to adapt to varying styles and content structures. Reference [17] explored this method by applying Transfer Learning techniques, leveraging out-of-domain essays to enhance model performance. They utilized two different datasets, ASAP and CRP. The authors claimed that they obtained state-of-the-art results and that their approach can improve the model's ability to generalize across diverse writing styles and topics.
- **Cross-Domain**
  This method assesses the model's ability to generalize by training it on one dataset and then validating its efficiency on a completely different dataset. Reference [29] argued that testing the model on a different dataset would enhance the generalizability across different domains. In their study, they used the public ASAP dataset, which focuses on English writing evaluation for grades 7-10, and the domain-specific OS dataset, which contains questions related to operating systems at the college level. Despite the differences in student level and context, the authors claimed that their model outperformed the baseline, demonstrating its robust cross-domain capabilities.

4) FAIRNESS

ML/DL models, especially pretrained models, have demonstrated powerful performance in AES tasks; however, such

TABLE 7. Summary of AES studies focused on holistic scoring.

Ref.	Model	Dataset	Metric	Application
[5]	BERT	ASAP	QWK: 0.79	Introduced R2BERT, a pretrained model using a multi-loss strategy that combines regression and ranking to enhance holistic scoring and overcome the limitations of standard BERT fine-tuning.
[28]	BERT/ LSTM-MoT	ASAP	QWK: 0.79	Proposed a multi-layer model combining LSTM-MoT for document-level features and BERT-LSTM for sentence-level features. The model was fed with shallow semantic features to enhance holistic scoring accuracy.
[36]	BERT	ASAP	QWK: 0.77	Combined manually extracted features, Word2Vec representation, and BERT embeddings. They argued that combining manual features and deep-learned features would enhance the scoring accuracy.
[42]	BERT	ASAP	QWK: 0.78	Fine-tuned BERT using three active learning strategies (Uncertainty-Based, Topological-Based, and Hybrid) to reduce the number of manually scored essays required for building accurate AES systems.
[47]	BERT	ASAP	QWK: 0.83	Employed MTL with fine-tuned BERT and attention mechanism to generate holistic scores for student responses.
[48]	BiLSTM/ RoBERTa	ASAP	QWK: 0.79	Used BiLSTM over a pretrained RoBERTa to address coherency issues and to enhance AES system scoring. Addressed the issue of long essay truncation (BERT 512 Token limitation) using BiLSTM to allow full essay analysis. The model also incorporated prompt text into the essay input to improve AES accuracy.
[51]	BERT	ASAP	QWK: 0.78	Proposed BERT model with extra adapters to enhance AES holistic scoring capabilities. Tried to reduce BERT training cost by freezing 99% of the trainable parameters.
[53]	BERT	NAEP	QWK: 0.78	Proposed an AES model based on multi-task and meta-learning to fine-tune BERT. The proposed model outperformed other AES systems. BERT was trained using passages, questions, and responses as combined inputs.
[55]	BERT	ASAP	Accuracy	To enhance scoring accuracy, the essay subject and a summarized version of the topic were inserted every 10 lines before fine-tuning BERT, helping the model learn the relationship between the essay topic.
[58]	BERT	ASAP (T1,2,7)	QWK: 0.79	Extracted topical information from the prompt using Automated Writing Evaluation, and then fed them with student responses during the fine-tuning of the BERT model to enhance scoring and improve model transparency.

TABLE 8. Summary of AES studies focused on trait scoring.

Ref.	Model	Dataset	Overall QWK	Traits QWK	Application
[29]	SBERT + LSTM	ASAP(T2) OS	QWK: 0.69 QWK: 0.67	Organization: 0.67 Word choice: 0.67	To capture coherence, each sentence is embedded individually using SBERT, and the embeddings are then fed to an LSTM model during training. The proposed model outperformed baselines in capturing coherence.
[37]	BERT/ Long-former	ASAP	QWK: 0.79	Grammar Style	Enhanced the AES scoring by evaluating the grammatical and stylistic factors to assess text quality. They demonstrated that the Longformer variant outperformed BERT due to its ability to handle longer essay input.
[47]	BERT	CELA	QWK: 0.78	Grammar: 0.87 Lexicon: 0.80 Global Org: 0.79 Local Org: 0.72 Support ideas: 0.74	Employed a Hierarchical model with Multi-Task Learning (MTL), fine-tuned BERT, and attention mechanisms to handle long essays and identify key parts that affect the final score. The use of the attention mechanism helped the model to focus on critical parts of the essay and capture essential information that influences scoring, thereby improving the overall scoring performance.
[57]	Long-former	-	QWK range 0.44 - 0.84	Main Idea: 0.78 Language: 0.84 Word Count: 0.44 Support: 0.55	A Longformer-based model was used to score various essay traits, grouped into relevant clusters. The model addresses challenges associated with scoring long essays.

TABLE 9. Summary of AES studies focused on cross-prompt and generalization.

Ref.	Model	Dataset	Metric	Application
[16]	BERT	ASAP (T2,3,4, 5,6)	QWK	Investigated the ability of BERT for cross-prompt generalization on the ASAP dataset. Compared the results of fine-tuning pretrained models against non-fine-tuned ones, and found that both approaches produced similar scoring performance.
[17]	BERT/ Longformer	ASAP, CRP	QWK	Used BERT for multi-scale essay representation and employed multiple losses and transfer learning approaches for out-of-domain essays. The model was trained on ASAP and generalized well on the CRP dataset.
[29]	SBERT + LSTM	ASAP, OS	QWK	A multi-task model combining SBERT embeddings with LSTM to handle sequential information. Authors argued that testing their model on a different dataset would enhance the generalizability across domains.
[52]	SBERT	ASAP	-	Classified essays as Good or Bad using the SBERT model in a cross-prompt setting.
[56]	BERT + NN	ASAP, TOEFL11	QWK	To improve generalization, they used PANN with disentangled representation learning with two separation strategies, quality-content and Quality-adherence. Utilized BERT and PA-net for representation learning.

models can inherit biases because they were pretrained on a large corpus of text in an unsupervised manner. This highlights the concerns regarding model fairness. AES systems should strive to score similar essays equally regardless of different aspects, such as race, gender, nationality, or socioeconomic background [16]. This principle ensures that similar essays written by different students from different groups are treated equally without discrimination.

TABLE 10. Summary of AES studies addressed fairness.

Ref.	Model	Dataset	Metric	Application
[16]	BERT	ASAP (T2,3,4,5,6)	QWK	Emphasized the importance of fairness in AES systems by ensuring that similar essays are scored equally, regardless of race, gender, nationality, or socioeconomic background
[50]	SBERT	ASAP (T7)	QWK	Investigated group fairness using cosine distance with SBERT embeddings. Essay pairs were compared using distance measures to ensure similar essays are treated similarly
[53]	In-context BERT	NEAP		Compared bias in models trained with and without students' authenticity information. Biases were found, but within an acceptable range

TABLE 11. Studies focusing on robustness against adversarial essays.

Ref.	Model	Dataset	Metric	Application
[45]	GloVe, ELMo, BERT	ASAP	QWK	Compared the performance of BERT, GloVe, and ELMo models on AES tasks to evaluate QWK and a custom robustness metric against adversarial input. BERT outperformed overall. However, fine-tuning improved QWK but reduced robustness, making the models more sensitive to lengthy text and lexical patterns.
[49]	CapsNet+ BERT	ASAP	QWK	Combined BERT with capsule networks and feature extraction to detect adversarial responses and assign them lower scores.
[51]	In-context BERT	NEAP	QWK	Fine-tuned BERT to detect adversarial responses. The model demonstrated high performance and outperformed the benchmark models.

Only a few studies have explored methods to assess and mitigate fairness bias in AES systems, as demonstrated in Table 10. Reference [50] investigated group fairness using cosine distance with SBERT embeddings and utilized the ASAP Prompt 7 dataset. This approach leverages the ability of the SBERT model to capture semantic similarity between pairs of essays. As a future study, the authors considered how individual fairness could contribute to group fairness. In addition, how can the AES system score essays submitted by students with special needs higher than normal students to appreciate their extra efforts?

Similarly, [53] explored fairness aspects such as student gender and origin in the AES system. They employed a BERT model on the NEAP dataset and calculated the average bias between each group's predicted and actual scores. They also compared the performance of two models, where the first model was trained with students' authenticity information included within the essay text, and the second model was trained without this information. They found unavoidable biases; however, these biases were within an acceptable range. They also suggested that fairness regulations should be included when training the ML models.

5) ROBUSTNESS

Model robustness refers to a model's ability to resist adversarial responses and cheating activities. In the AES context, adversarial responses are essays that appear to be well-written to human readers but are intentionally designed to confuse or manipulate the AES system. Ensuring the model's robustness against such attacks is crucial for ensuring the system's reliability and robustness. Table 11 summarizes key studies that have explored the robustness of AES models.

Out of these studies, [45] compared the performances of various models and found that BERT achieved the highest robustness. Interestingly, they also found that the models tended to assign higher scores to essays with longer sentences. Furthermore, higher scores were assigned to essays in which common words were frequently used. These findings highlight a potential vulnerability that can be exploited by incorporating longer sentences with frequently used semantically irrelevant words in adversarial attacks.

Similarly, [49] and [51] investigated whether AES models are vulnerable to adversarial attacks and proposed novel detection methods. Reference [49] combined BERT-enhanced capsule networks with feature extraction to detect and score adversarial responses lower. In contrast, [51] fine-tuned a BERT model using the ASAP dataset and then tested the model for detecting adversarial responses. Both studies claimed that their model performed well against adversarial attacks and outperformed other benchmark models in terms of scoring.

V. DISCUSSION

This section discusses the findings obtained from the systematic review, with a focus on key challenges and limitations identified in the reviewed papers. The discussion is intended to address Research Question 3 (RQ3.

The AES research landscape is rich, with many studies exploring various areas. The primary focus remains on improving the scoring process by minimizing the gap between human scores (i.e., the gold standard) and machine-generated scores. However, many studies have addressed other concerns that emerged during this extensive research journey, including limited training datasets, model generalizability, model explainability, fairness in scoring, and the ability to provide meaningful feedback to students.

The advancement of ML techniques and AES has progressed symbiotically. Starting with the traditional statistical models that laid the groundwork for AES research. These models were labor-intensive due to the need for manual feature engineering [26]. The introduction of NN/DL models, such as CNNs and DNNs, provided significant advancements because of their ability to learn complex patterns and eliminate the need for manual feature engineering. Among these, the LSTM model gained popularity in AES research due to its ability to process input text sequentially. However, challenges remain, including the closed-box nature of these models, the need for intensive computational powers, and large datasets for training [29].

Transformers have revolutionized NLP by enabling parallel processing of text and capturing word context through the self-attention mechanisms. Unlike sequential NN models such as RNNs and CNNs, Transformers can capture the relationship between words across entire sentences as well as the whole document efficiently. Typically, they use an encoder-decoder architecture. The encoder generates a contextualized representation of the input text, while the decoder uses these representations to produce output. The self-attention mechanism in both components helps the model to focus on the most relevant parts in the text sequences [12], [13], [14].

There are many pretrained transformer-based models such as BERT, GPT, and T5. These models are trained on massive corpora in an unsupervised manner to learn the general language patterns, which makes them highly efficient in handling complex language tasks such as AES [45]. Furthermore, these models can be fine-tuned on domain-specific datasets to allow them to adapt and improve the output accuracy for specific tasks [5].

The utilization of Transformers pretrained models has significantly advanced the AES research by providing robust models that could understand and evaluate student essays and provide scoring more contextually. BERT has emerged as the most used pretrained model in AES research, as evident in the results presented in the previous section. This is due to its strong performance in various NLP tasks in general, and AES in particular. It utilizes only the encoder part of the transformer, which allows the model to understand the context bidirectionally. Notably, the highest QWK of 0.83 was achieved by [47] using a model that employed BERT. However, several challenges remain with Transformers pretrained models, including their closed-box nature, high computational requirements, longer training times, and the need for large datasets [16]. Additionally, issues such as data quality, biases in training data, generalization, and overfitting still pose significant obstacles that need to be addressed.

The primary focus of AES research is leveraging advancements in AI and ML/DL techniques to generate human-like scores. AES scoring is typically categorized into holistic and trait scoring. Holistic scoring requires the model to generate a single overall score for an essay. Recent studies have shown promising results in this area, with models achieving high QWK accuracy. On the other hand, trait scoring focuses on assigning separate scores for various aspects of an essay, such as concept, organization, grammar, and context, which can be used later to provide feedback to students and identify areas for improvement.

Although current research trends show that the use of Transformers pretrained models in AES tasks such as trait scoring and cross-prompt generalization is underutilized with limited performance, these areas hold potential and should not be overlooked. Continued research that incorporates the latest ML models, along with insights from previous studies, is expected to drive further advancements in AES. Additionally, the number of AES studies that utilized Transformer-based pretrained models (e.g., BERT) for tasks like trait scoring and cross-prompting are relatively lower than those employing NN/DL models (e.g., CNN and RNN models). Despite NN models achieving acceptable performance in trait scoring, their results still fall short compared to holistic scoring. Replicating existing NN-based AES studies, such as [7], [34], [35], and [59] with pretrained models could be a promising area for future research.

Another significant challenge in AES research is the lack of specialized datasets. Most AES studies rely on the ASAP dataset as the primary benchmark, highlighting the limited options available. Additionally, most publicly available datasets are tailored toward English-language assessment, e.g., TOEFL11, CELA, or the Cambridge English assessment datasets. The lack of datasets is understandable because of the labor-intensive process of creating them, which requires extensive manual work and human expertise, as evidenced by [29] in their attempt to develop the Operating Systems essays dataset. Furthermore, this limitation is also noticed in commercial products, which primarily focus on English-language assessments and feedback. However, the potential of AES should extend beyond English-language assessments with sufficient adaptability to handle diverse domains such as medicine, engineering, and computer science. The power of Transformers pretrained models could partially address this problem. These powerful models can be fine-tuned with relatively small datasets to cover new areas with less domain-specific data. This approach would reduce the need for specialized datasets and open doors for broader applications of AES systems [47].

Fine-tuning pretrained models is a common method for enhancing their ability to handle specific tasks. Nevertheless, this remains a topic of debate between AES researchers. On one hand, [16] have suggested that base pretrained models can achieve results closely similar to the fine-tuned ones. On the other hand, [47] have demonstrated that fine-tuning pretrained models can lead to considerable improvements in QWK scores. However, fine-tuning pretrained models usually requires increased computational power and longer training times. Therefore, further research is required to establish the best practices for fine-tuning pretrained models while considering the relevant tradeoffs.

The input length limitation in BERT, capped at 512 tokens, is another challenge. This limitation may lead to truncating longer essays with loss of valuable information, which could affect scoring accuracy. This issue has been addressed through various methods such as adding CNN or RNN layers to the model, using data augmentation techniques to summarize long essays, or employing Longformer architectures to handle longer text input.

In addition to the above challenges, pretrained models may struggle to capture the overall essay context because they are typically trained on sentence-level datasets. Reference [17] proposed a multilayer architecture incorporating an RNN layer with BERT to overcome this challenge and enhance the model's performance. Moreover, the closed-box nature of DL models remains a major challenge. Understanding how the model arrives at the generated score remains an ongoing area of research, where Explainable AI (XAI) and trait scoring may offer a partial solution.

Overall, transformer-based models, particularly pretrained models like BERT, hold extensive potential for AES tasks. However, overcoming the above limitations is crucial for unlocking their full potential and achieving broader adoption

VI. RECOMMENDATIONS AND FUTURE WORK

This section addresses Research Question 4 (RQ4) by outlining promising directions for advancing AES research. The review of relevant studies revealed significant opportunities and highlights key directions for future investigation, including:

- Broadening and extending the scope of the current research to include a wider range of studies, such as conference papers, to capture emerging trends and methods in AES. Furthermore, explore the research on Automated Writing Scoring (AWS), which focuses on providing feedback to improve students' writing skills.
- Exploring the performance of other pretrained models, as BERT is predominant in most of the shortlisted studies. Future research should focus on the use of other pretrained models, such as XLNet, GPT, and other models, to determine if they offer better performance in AES tasks.
- Revisiting the traditional NN/DL models that utilized LSTM or RNN, particularly those with publicly available code, and rebuilding them by replacing the NN models with the advanced pretrained models. This approach could reveal potential improvements in scoring accuracy.
- Enhancing both Holistic and trait scoring remains a primary objective. The focus remains on improving both holistic and trait scoring methods, exploring how they can be integrated to refine overall scoring approaches.
- Focusing on further research to enhance the generalizability of AES systems across different prompts, domains, and genres.
- Advancing the model explainability to address the closed-box nature of ML/DL models.
- Aligning with the emerging trends to keep AES research relevant and innovative. For example, exploring how advanced LLMs, such as OpenAI GPT, Falcon LLM, and LLaMA, can enhance AES scoring accuracy and expand the applicability to feedback generation.

By exploring these promising areas, more potential can be unlocked in AES research, leading to more robust models that can produce more accurate scores and empower students with detailed feedback to improve their writing skills.

VII. CONCLUSION

This systematic review explored the current landscape of AES research, with a particular focus on the utilization of Transformers-based models, specifically the pretrained models, by reviewing relevant literature from the last five years. This study followed the PRISMA guidelines to select, screen, and shortlist studies that fell within the defined scope. After a thorough analysis of the selected papers, data synthesis revealed five main areas of AES research: improving holistic scoring, trait scoring, cross-prompt and model generalizability, model fairness, and robustness against adversarial attacks.

Furthermore, the study has briefly outlined the evolution of ML approaches in AES research over time, from traditional statistical models and DL architectures to the more recent adoption of Transformer-based pretrained models. The analysis of AES studies that used Transformers pretrained models revealed a clear trend in using the BERT model (or one of its variants) due to their effectiveness in handling complex NLP tasks. However, the analysis also highlighted the challenges associated with these advanced models, including computational costs, input length restrictions, and issues related to the closed-box nature and model explainability.

Despite these challenges, the potential for transformer-based pretrained models in AES research is substantial. The ability of these models to process input sequences both in parallel and bidirectionally, while capturing context across longer text, has proven to be advantageous in enhancing scoring accuracy. Researchers are actively exploring new approaches to leverage these models to improve scoring accuracy, better generalizability, and enhanced feedback. The next logical step is to explore the recent trends in NLP, particularly the integration of Generative AI (GenAI) and large language models (LLMs) like OpenAI GPT, to further enhance AES capabilities. By addressing the current limitations and embracing emerging technologies, more efficient, robust, transparent, and insightful AES systems can be developed.

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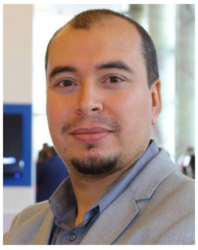

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